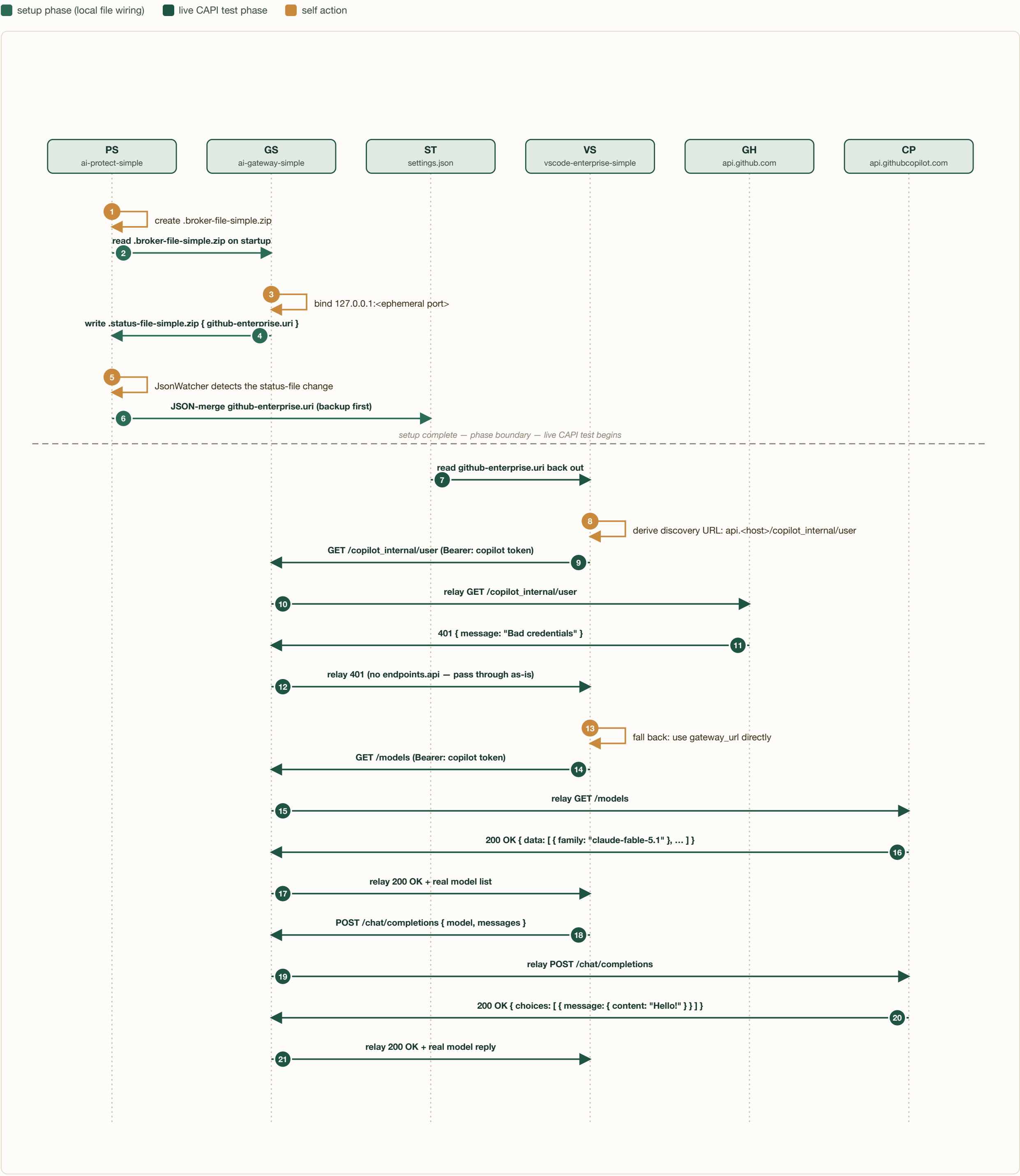


3.2.1. Full sequence, as actually run — every object, every message

The live end-to-end test of the github-enterprise.uri CAPI-discovery hijack mechanism, using the standalone ai-gateway-simple / ai-protect-simple / vscode-enterprise-simple tools against real api.github.com and api.githubcopilot.com traffic.



Step-by-step

- 1

ai-protect-simple writes .broker-file-simple.zip PS (self)
A minimal, real `BrokerFile` naming one agent, `vscode-copilot` — the same wire type the real product uses, not a reimplementation.
- 2

ai-gateway-simple reads it on startup PS → GS
Confirms it should be listening on behalf of that agent before it binds anything.
- 3

ai-gateway-simple binds an ephemeral local port GS (self)
127.0.0.1:<port> — nothing is exposed off-box.
- 4

Writes .status-file-simple.zip GS → PS
A real `StatusFile` naming the resolved `github-enterprise.uri` in a `ConfigDelta`, for ai-protect-simple to pick up.
- 5

ai-protect-simple's JsonWatcher detects the change PS (self)
Deliberately renamed control files (`.broker-file-simple.zip` / `.status-file-simple.zip`) so this can never collide with a real ai-protect/ai-gateway pair running on the same machine.
- 6

JSON-merge into the real settings.json PS → ST
Backs the file up once before the first write; every test run described here was fully reverted afterward via `--revert`.
- 7

vscode-enterprise-simple reads github-enterprise.uri back out ST → VS
The same key a real VS Code Copilot Chat install would read — confirms the merge in step 6 actually landed.
- 8

Derives the discovery URL VS (self)
VS Code's own exact formula, traced from `defaultAccount.ts`: prepend `api.` onto the configured hostname, unconditionally.
- 9

GET /copilot_internal/user VS → GS
The genuine discovery call, sent with a real Copilot bearer token (from the OAuth device flow) and the real client headers (User-Agent, Editor-Version, Editor-Plugin-Version).
- 10

Relay to the real api.github.com GS → GH
ai-gateway-simple makes a real, direct outbound call — this is the test tool proving the mechanism, not the shipped listener's actual path (see §4 of the write-up).
- 11

401 Bad credentials GH → GS
Expected. This bearer token is the Copilot-specific one from the `/copilot_internal/v2/token` exchange, not the raw GitHub OAuth token discovery wants — a known token-type mismatch, not a failure of the relay. **Caveat: this means a real, successful discovery response — a 200 with a genuine endpoints.api field actually being rewritten — was never observed live in this test.** The rewrite itself is confirmed by reading the source (`githubEndpoints.ts`, the `copilotApiService.test.ts` tests), not by watching it happen here; only the relay's forward-and-pass-through-on-failure behavior was exercised live.
- 12

Relay the 401 back untouched GS → VS
There's no `endpoints.api` field in a 401 to rewrite, so it passes through as-is.
- 13

Client falls back to the gateway URL directly VS (self)
Exactly what it would do if discovery had succeeded and named this same gateway as the CAPI host.
- 14

GET /models VS → GS
A real, authenticated listing call — the exact shape of request an enterprise-flagged VS Code would make right after a successful discovery.
- 15

Relay to the real api.githubcopilot.com GS → CP
Forwarded with the same real bearer token and client headers.
- 16

200 OK, a real ~52KB model list CP → GS
Genuine capability metadata for real models (including claude-fable-5.1). A listing endpoint, though — not proof that a completion works.
- 17

Relay the model list back GS → VS
Round-trips intact through the gateway relay.
- 18

POST /chat/completions VS → GS
The actual proof step: a real inference request, same payload shape and headers already validated directly against real CAPI by `copilot_chat_call.sh`.
- 19

Relay to the real api.githubcopilot.com GS → CP
Same relay path as step 15, now carrying a generation request instead of a listing.
- 20

200 OK, a genuine reply from a real model CP → GS
`gpt-4o-2024-11-20` replied "Hello!" to the test prompt, with real token accounting (`prompt_tokens: 14`, `completion_tokens: 2`).
- 21

Relay the real model reply back GS → VS
Comes back through the gateway intact — the mechanism is now proven with a genuine, generated completion, not just a listing.